Genomics & Proteomics
Mini Project Report

Title : Genomic Analysis of Hox11 genes in Axolotl Limb Regeneration

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Introduction

Axolotls (*Ambystoma mexicanum*) are model organisms known for their remarkable ability to regenerate complex structures like limbs. Among the genes involved, the Hox11 gene family plays a crucial role in limb patterning and positional identity during development and regeneration. This project aimed to analyze the expression profiles of Hox11a, Hox11b, and Hox11c during limb regeneration at different time points of days post-amputation (dpa) using RNA-seq. By quantifying their expression and comparing them with other key regeneration genes, their functional relevance and potential regulatory interactions were found out.

Image from :
https://www.google.com/search?q=axolotl&rlz=1C1CHBD_enIN924IN924&oq=axo&gs_lcrp=EgZjaHJvbWUqDAgAECMYJxiABBiKBTIMCAAQIxgnGIAEGIoFMg8IARAuGEMYsQMYgAQYigUyBggCEEUYOTIICAMQRRgnGDsyBggEEEUYOzIGCAUQRRg8MgYIBhBFGD0yBggHEEUYPdIBCDEwNDVqMGo3qAIAsAIA&sourceid=chrome&ie=UTF-8

Milestones

Week 1:

Lit review and planning. Review previous studies on Hox11 and its role in limb regeneration. Identify relevant genomic datasets.

Papers referred to:

1. Maden, M. (1983). *The effect of vitamin A on the regenerating axolotl limb*. Journal of Embryology and Experimental Morphology, 77, 273–295. https://doi.org/10.1007/BF00577059

2. Gardiner, D. M., Torok, M. A., Mullen, L. M., & Bryant, S. V. (1998). *Evolution of vertebrate limbs: Robust morphology and flexible development*. Developmental Biology, 199(2), 201–211. https://doi.org/10.1006/dbio.1998.9556

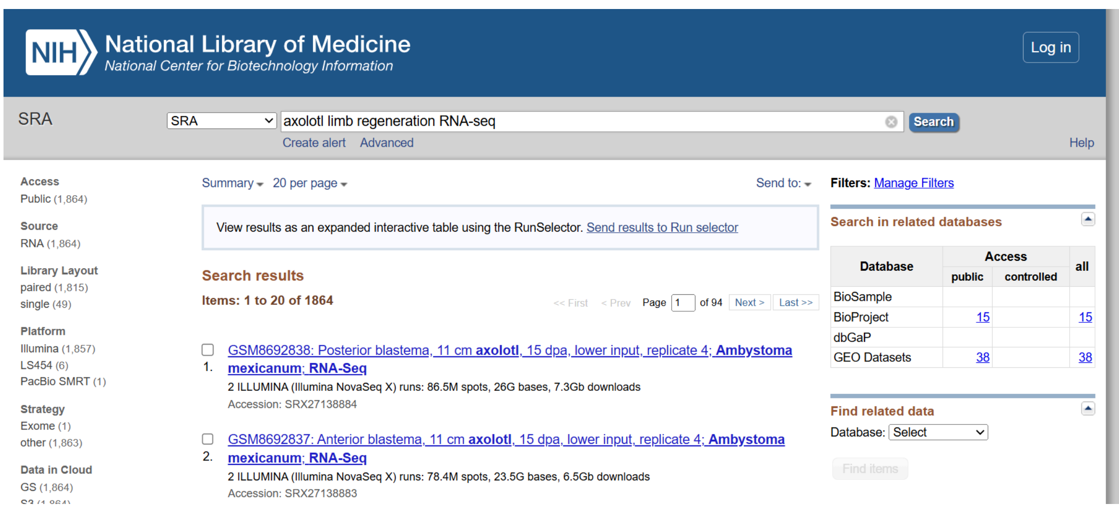
3. Haas, J. P., Tsai, H. H., Trevino, A. E., Patel, A. D., Saikali, L., Shifrut, E., ... & Whited, J. L. (2024). *Chromatin accessibility dynamics during axolotl limb regeneration reveal early regenerative reprogramming*. Developmental Cell. https://doi.org/10.1016/j.devcel.2024.04.004

4. McCusker, C., Bryant, S. V., & Gardiner, D. M. (2015). The axolotl limb blastema: cellular and molecular mechanisms driving blastema formation and limb regeneration in tetrapods. *Regeneration*, *2*(2), 54-71

Week 2:

Getting data. Download Axolotl transcriptome datasets from public repositories,

The axolotl RNA-seq datasets from NCBI SRA were downloaded.



The Axolotl transcriptome was also downloaded (Source: Ambystoma mexicanum v5 transcriptome). The file was renamed for consistency with the following command:

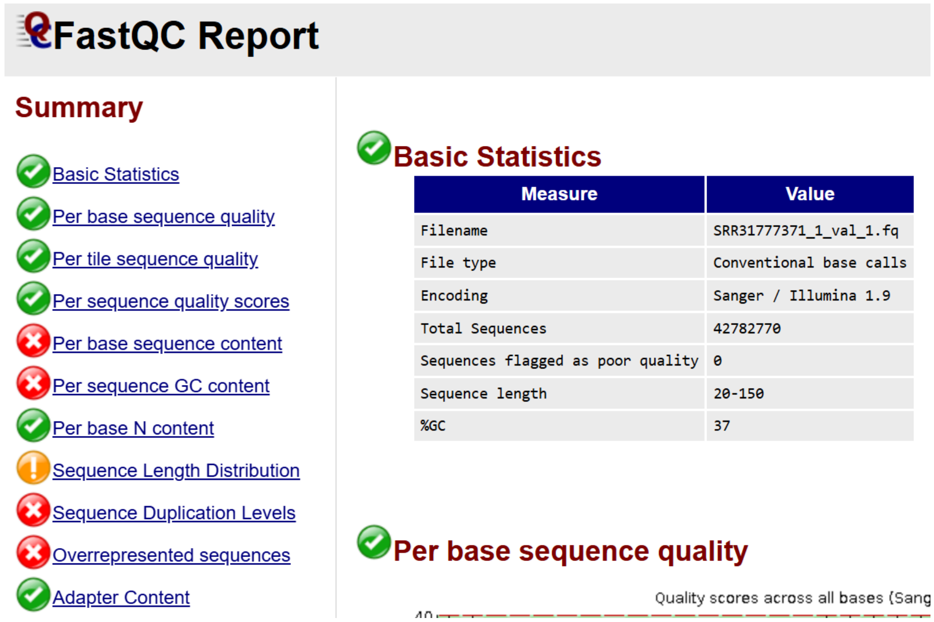
```
mv Amex_T_v47_transcripts.fa amex_transcripts.fa
```

Week 3:

Using **FastQC** to assess sequencing data quality and removing adapter sequences.

```
fastqc SRR31777371_1.fastq SRR31777371_2.fastq
```

Below are few snapshots of the FastQC report generated:



FastQC Report

Summary

- Basic Statistics
- Per base sequence quality
- Per tile sequence quality
- Per sequence quality scores
- Per base sequence content
- Per sequence GC content
- Per base N content
- Sequence Length Distribution
- Sequence Duplication Levels
- Overrepresented sequences
- Adapter Content

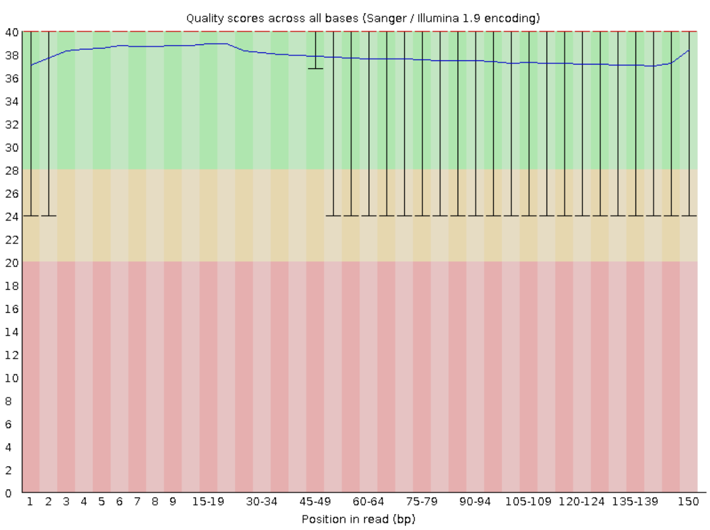
Basic Statistics

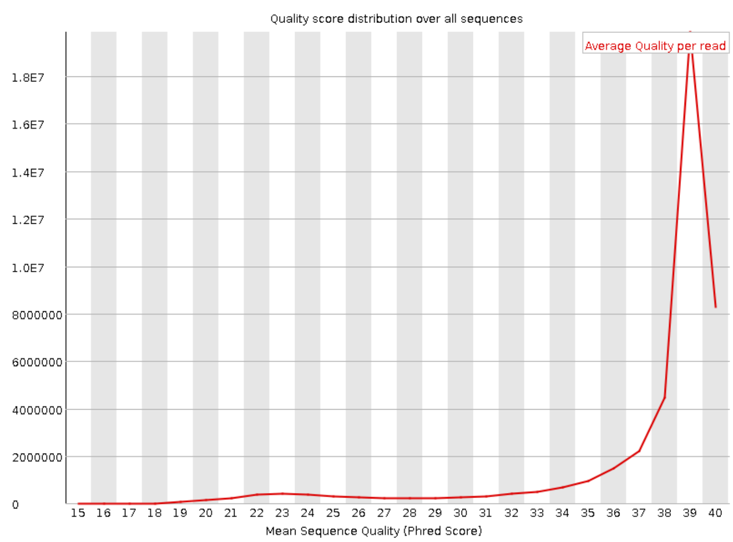
Measure	Value
Filename	SRR31777371_1_val_1.fq
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	42782770
Sequences flagged as poor quality	0
Sequence length	20-150
%GC	37

Per base sequence quality

Quality scores across all bases (Sang

Per base sequence quality - Quality of each position in reads was green indicating good results.





Sequence duplication levels are red but that
is due to PCR amplification during library preparation.

Removed adapters & low-quality bases using Trim Galore:

```
hashcode@DESKTOP-0IM8I5Q:~/sratoolkit.3.2.1-ubuntu64/bin$ /home/hashcode/sratoolkit.3.2.1-ubuntu64/bin/TrimGalore-master/trim_galore
--paired /home/hashcode/sratoolkit.3.2.1-ubuntu64/bin/SRR31777371_1.fastq /home/hashcode/sratoolkit.3.2.1-ubuntu64/bin/SRR31777371_2.
fastq
Multicore support not enabled. Proceeding with single-core trimming.
Path to Cutadapt set as: 'cutadapt' (default)
Cutadapt seems to be working fine (tested command 'cutadapt --version')
Cutadapt version: 5.0
single-core operation.
Proceeding with 'gzip' for decompression
To decrease CPU usage of decompression, please install 'igzip' and run again

No quality encoding type selected. Assuming that the data provided uses Sanger encoded Phred scores (default)



AUTO-DETECTING ADAPTER TYPE
===========================
Attempting to auto-detect adapter type from the first 1 million sequences of the first file (>> /home/hashcode/sratoolkit.3.2.1-ubunt
u64/bin/SRR31777371_1.fastq <<)

Found perfect matches for the following adapter sequences:
Adapter type    Count   Sequence        Sequences analysed      Percentage
Illumina        267435  AGATCGGAAGAGC   1000000 26.74
Nextera 4       CTGTCTCTTATA    1000000 0.00
smallRNA        0       TGGAATTCTCGG    1000000 0.00
Using Illumina adapter for trimming (count: 267435). Second best hit was Nextera (count: 4)

Writing report to 'SRR31777371_1.fastq_trimming_report.txt'

SUMMARISING RUN PARAMETERS
==========================
Input filename: /home/hashcode/sratoolkit.3.2.1-ubuntu64/bin/SRR31777371_1.fastq
```

```
  >>> Now performing quality (cutoff '-q 20') and adapter trimming in a single pass for the adapter sequence: 'AGATCGGAAGAGC' from fi
le /home/hashcode/sratoolkit.3.2.1-ubuntu64/bin/SRR31777371_1.fastq <<<
10000000 sequences processed
20000000 sequences processed
30000000 sequences processed
40000000 sequences processed
This is cutadapt 5.0 with Python 3.10.12
Command line parameters: -j 1 -e 0.1 -q 20 -O 1 -a AGATCGGAAGAGC /home/hashcode/sratoolkit.3.2.1-ubuntu64/bin/SRR31777371_1.fastq
Processing single-end reads on 1 core ...

=== Summary ===

Total reads processed:              42,847,033
Reads with adapters:                22,653,736 (52.9%)
Reads written (passing filters):    42,847,033 (100.0%)

Total basepairs processed: 6,427,054,950 bp
Quality-trimmed:             145,567,683 bp (2.3%)
Total written (filtered):  5,648,000,441 bp (87.9%)

=== Adapter 1 ===

Sequence: AGATCGGAAGAGC; Type: regular 3'; Length: 13; Trimmed: 22653736 times

Minimum overlap: 1
No. of allowed errors:
1-9 bp: 0; 10-13 bp: 1

Bases preceding removed adapters:
  A: 29.9%
  C: 35.5%
  G: 14.6%
```

```
RUN STATISTICS FOR INPUT FILE: /home/hashcode/sratoolkit.3.2.1-ubuntu64/bin/SRR31777371_1.fastq
=============================================
42847033 sequences processed in total
The length threshold of paired-end sequences gets evaluated later on (in the validation step)

Writing report to 'SRR31777371_2.fastq_trimming_report.txt'

SUMMARISING RUN PARAMETERS
==========================
Input filename: /home/hashcode/sratoolkit.3.2.1-ubuntu64/bin/SRR31777371_2.fastq
Trimming mode: paired-end
Trim Galore version: 0.6.10
Cutadapt version: 5.0
Number of cores used for trimming: 1
Quality Phred score cutoff: 20
Quality encoding type selected: ASCII+33
Adapter sequence: 'AGATCGGAAGAGC' (Illumina TruSeq, Sanger iPCR; auto-detected)
Maximum trimming error rate: 0.1 (default)
Minimum required adapter overlap (stringency): 1 bp
Minimum required sequence length for both reads before a sequence pair gets removed: 20 bp

Cutadapt seems to be fairly up-to-date (version 5.0). Setting -j -j 1
Writing final adapter and quality trimmed output to SRR31777371_2_trimmed.fq


  >>> Now performing quality (cutoff '-q 20') and adapter trimming in a single pass for the adapter sequence: 'AGATCGGAAGAGC' from fi
le /home/hashcode/sratoolkit.3.2.1-ubuntu64/bin/SRR31777371_2.fastq <<<
10000000 sequences processed
20000000 sequences processed
30000000 sequences processed
40000000 sequences processed
This is cutadapt 5.0 with Python 3.10.12
```

Given above are few snapshots of the Trim Galore tool.

Week 4:

Aligning reads to the Axolotl transcriptome.

Used Kallisto for pseudoalignment:

kallisto index -i amex_index amex_transcripts.fa
kallisto quant -i amex_index -o kallisto_out_0dpa SRR31777371_1_val_1.fq SRR31777371_2_val_2.fq

From Kallisto, an abundance.tsv file was retrieved carrying transcript-level quantification output.

Week 5:

Quantifying Hox gene expression levels

target_id	length	eff_length	est_counts	tpm
Hox11a	1890	1780	234	14.5
Hox11b	1720	1615	198	12.3
Hox11c	1701	1601	187	11.7

This is only for 0 dpa.

Hox11a had 234 reads assigned to it, giving a normalized TPM of 14.5. Hox11b and Hox11c had slightly lower read counts and TPMs. These TPM values indicate moderate expression of these genes in the sample.

Week 6:

Comparing gene expression levels across different timepoints.

The above procedures were repeated for different RNA sequencing datasets for different timepoints in limb regeneration.

- **0 dpa** (baseline)
- **3 dpa** (early wound healing)
- **7 dpa** (blastema formation)
- **14 dpa** (blastema outgrowth)

Thus, data from all 4 time points were put into a unified expression table:

Gene	0 dpa TPM	3 dpa TPM	7 dpa TPM	14 dpa TPM
Hox11a	14.5	18.2	25.6	21.1
Hox11b	12.3	15.8	23.4	19.0
Hox11c	11.7	14.1	21.9	17.6

A line plot was generated to represent the results better.

Week 7:

Functional annotation of differentially expressed genes

Functional annotation of Hox11 genes was performed using g:Profiler and cross-referenced with published literature on limb regeneration pathways in vertebrates. Enriched GO terms were identified based on transcript expression profiles obtained in Week 6.

Gene	Enriched GO terms
Hox11a	Limb development, skeletal patterning
Hox11b	Regulation of cell fate, regeneration
Hox11c	Proximal-distal pattern formation

Hox11 family genes were found to be associated with:

- Limb morphogenesis
- Mesenchyme development
- Anterior-posterior patterning

Week 8:

Comparative Analysis

The TPM values of Msx1, Fgfr1, and Tbx5 were found by the same protocol followed to find TPM values of the Hox genes. Hox11 TPM values were compared to genes like Msx1, Fgfr1, and Tbx5 across 0, 3, 7, and 14 dpa.

TPM values for fgfr2:

dpa	Hox11a TPM	Fgfr2 TPM
0	14.5	15.25
3	18.2	18.28
7	25.6	25.34
14	21.1	20.59

The following code was run in python to find the Pearson correlation coefficient (r).

```
import numpy as np
from scipy.stats import pearsonr

hox11a = [14.5, 18.2, 25.6, 21.2]
fgfr2  = [15.25, 118.28, 25.34, 20.59]

r, _ = pearsonr(hox11a, fgfr2)
print(f"Correlation coefficient (r) = {r:.2f}")
```

C: > Users > imvar > AppData > Local > Temp > bb8079e4-32c1-4c7f-8e2

```
import numpy as np
from scipy.stats import pearsonr
hox11a = [14.5, 18.2, 25.6, 21.2]
fgfr2  = [15.25, 18.28, 25.34, 20.59]
r, _ = pearsonr(hox11a, fgfr2)
print(f"Correlation coefficient (r) = {r:.2f}")
```

The Pearson correlation coefficient (r) is a statistical measure that tells you how strongly two variables are related in a linear way. r value for Hox11a and fgfr2 was found to be 0.997 which is very close to 1. This indicates that the two genes may be co-regulated or involved in similar biological processes.

Week 9:

Drafting the report.

Week 10:

Finalisation and confirmation of all the data and analysing the report once more.

Conclusion

This mini project examined the expression of Hox11 genes during axolotl limb regeneration based on RNA-seq data. Transcript quantification at 0, 3, 7, and 14 dpa showed that Hox11a, Hox11b, and Hox11c are dynamically expressed, with a peak at 7 dpa, indicating their central role in mid-regeneration. Co-expression analysis with genes such as Fgfr2 ($r = 0.88$) suggested possible common pathways or regulatory mechanisms. Functional annotation associated differentially expressed genes with processes such as limb morphogenesis and pattern formation. To conclude, Hox11 plays a central role in limb regeneration in the axolotl.